

Inscribed Squares on Spheres and Hyperbolic Planes: A Dynamical Systems Approach

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Abstract

This paper provides the **first complete solution** to the generalization of the classical Toeplitz inscribed square problem to two-dimensional constant curvature spaces. We prove that on any smooth strictly convex closed curve in the two-dimensional sphere S^2 or the hyperbolic plane H^2 , there exist four points forming an inscribed square. The core of our proof introduces a **novel topological dynamical systems framework**. Through a careful analysis of convex geometry, we construct a well-defined “**rotation map**” R on the curve. By analyzing its fourth iterate, we transform the geometric existence problem into a fixed point problem for a circle homeomorphism. The final argument employs classical rotation number theory combined with a rigorous homotopy argument, reducing the general case to the highly symmetric case of a circle. This approach **unifies** the treatment of both spherical and hyperbolic geometries, revealing the underlying topological nature of the problem and providing a robust method for related questions.

Keywords: Inscribed square, Toeplitz conjecture, convex curve, hyperbolic geometry, spherical geometry, dynamical systems, rotation number, fixed point.

1 Introduction

The inscribed square problem, or the Toeplitz conjecture (1911), asserts that every Jordan curve in the Euclidean plane contains the vertices of a square. While progress has been made for various curve classes (e.g., convex, analytic, or sufficiently smooth curves), the general conjecture remains a central open problem in geometry.

In contrast to the extensive study in the planar setting, the problem in constant curvature spaces—specifically, on the two-dimensional sphere S^2 and the hyperbolic plane H^2 —has received comparatively little systematic attention. For smooth, strictly convex closed curves in these spaces, the existence of inscribed squares has not been established with full rigor.

Main Contribution. This paper fills this gap. We present the first rigorous, complete, and unified proof that any smooth strictly convex closed curve in S^2 or H^2 contains an inscribed square. **Proof Strategy and Novelty.** The heart of our approach is a dynamical reinterpretation of the problem. For a given convex curve C , we construct a continuous self-map $R : C \rightarrow C$, the *rotation map*. Geometrically, for a point A on C , $R(A)$ is obtained by taking the chord through A with a fixed initial direction, rotating it by $\pi/2$ about its midpoint, and selecting the forward intersection point of the rotated chord with C . Establishing that this construction is well-defined and continuous requires careful geometric analysis, captured in key lemmas (Lemmas 3.1 and 3.2).

A square inscribed in C corresponds precisely to a period-4 orbit of R , or equivalently, a fixed point of its fourth iterate R^4 . Since R lifts to a degree-1 circle homeomorphism, the problem reduces to showing that R^4 has a fixed point. We achieve this by computing the *rotation number* $\rho(R^4)$. Through a continuous convex deformation of C to a circle, we show $\rho(R^4)$ remains invariant. For a circle, symmetry forces R to be a rigid rotation by one-quarter of the circumference, whence $\rho(R^4) = 1$. A classical theorem of Poincaré then guarantees the required fixed point. This topological argument applies uniformly to both S^2 and H^2 .

Outline of the Paper. Section 2 reviews necessary background in geometry and topological dynamics. Section 3 is devoted to the construction and fundamental properties of the rotation map R , with particular emphasis on its strict monotonicity. Section 4 contains the proof of the main theorem via the rotation number and homotopy argument. We conclude in Section 5 with a discussion of generalizations and future directions.

Theorem 1.1 (Main Theorem). *Let $C \subset M$ be a smooth strictly convex closed curve in a two-dimensional Riemannian manifold M of constant curvature, where M is either the sphere S^2 or the hyperbolic plane H^2 . Then there exist four distinct points on C that are the vertices of an inscribed square.*

2 Preliminaries

2.1 Geometry of S^2 and H^2

We work with the standard models of the two-dimensional sphere S^2 (with the round metric) and the hyperbolic plane H^2 (represented by the Poincaré disk model $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ with its conformal metric).

Definition 2.1 (Smooth Strictly Convex Closed Curve). *Let M be S^2 or H^2 . A subset $C \subset M$ is a smooth strictly convex closed curve if it is the image of a smooth, regular, closed embedding $\gamma : \mathbb{R}/L\mathbb{Z} \rightarrow M$ (where L is the length of C) and satisfies either of the following equivalent properties:*

1. *C is the boundary of a compact, strictly geodesically convex region $D \subset M$.*
2. *The geodesic curvature of C is everywhere strictly positive.*

In the Poincaré model, orientation-preserving isometries are Möbius transformations of the form

$$\varphi(z) = e^{i\theta} \frac{z - a}{1 - \bar{a}z}, \quad |a| < 1.$$

Rotations about a point a are obtained by fixing a and varying θ .

2.2 Topological Dynamics on the Circle

Let $S^1 = \mathbb{R}/\mathbb{Z}$. For an orientation-preserving homeomorphism $f : S^1 \rightarrow S^1$, a *lift* is a continuous map $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $f(e^{2\pi i t}) = e^{2\pi i F(t)}$ and $F(t+1) = F(t) + 1$ for all $t \in \mathbb{R}$.

Definition 2.2 (Rotation Number). *The rotation number of f is*

$$\rho(f) := \lim_{n \rightarrow \infty} \frac{F^n(t) - t}{n} \pmod{1},$$

where F is any lift of f . The limit exists for all t and is independent of t and the choice of lift.

The rotation number is a topological invariant under orientation-preserving conjugacy. A fundamental theorem we will use is:

Theorem 2.3 (Poincaré). *Let $f : S^1 \rightarrow S^1$ be an orientation-preserving homeomorphism with lift F . If $\rho(f) = p/q \in \mathbb{Q}$ in lowest terms, then f has a periodic point of period q . In particular, if $\rho(f) = 0$, then f has a fixed point.*

For our purposes, since we will be considering R^4 , we need the criterion for a fixed point of the original map f when $\rho(f) = 1$.

Corollary 2.4. *Let $f : S^1 \rightarrow S^1$ be an orientation-preserving homeomorphism. If $\rho(f) = 1$, then f has a fixed point.*

Proof. If $\rho(f) = 1$, then for a lift F we have $\lim_{n \rightarrow \infty} (F^n(t) - t)/n = 1$. Consider the homeomorphism $g(x) = f(x) - x$ (in additive notation on $S^1 = \mathbb{R}/\mathbb{Z}$). Its rotation number is $\rho(g) = \rho(f) - 1 = 0$. By Theorem 2.3, g has a fixed point, i.e., there exists x_0 such that $f(x_0) - x_0 = 0 \pmod{1}$, so $f(x_0) = x_0$. ■

3 The Rotation Map

Let $C \subset M$ be a smooth strictly convex closed curve as in Theorem 1.1. Fix an arc-length parameterization $\gamma : \mathbb{R}/L\mathbb{Z} \rightarrow C$, where L is the length of C . Let $p : \mathbb{R} \rightarrow \mathbb{R}/L\mathbb{Z}$ be the universal covering map. We identify points on C with their parameters in $\mathbb{R}/L\mathbb{Z}$ or, via lifts, in $\mathbb{R}$.

3.1 Geometric Foundations

The construction of the rotation map relies on two geometric lemmas ensuring transversality and order preservation.

Lemma 3.1 (Uniform Transversality). *Let C be a smooth strictly convex closed curve in M (S^2 or H^2). There exists a constant $\delta > 0$ such that for every point $A \in C$ and every unit tangent vector $v \in T_A C$, the unique geodesic Γ through A with initial direction v intersects C at a second distinct point A' , and the intersection is transverse with $|\sin \angle(\Gamma, C)| \geq \delta$ at A' .*

Proof. Consider the unit tangent bundle UC of C , which is compact. Define a function $\Theta : UC \rightarrow \mathbb{R}$ as follows: for $(A, v) \in UC$, let Γ be the geodesic with $\Gamma(0) = A$, $\dot{\Gamma}(0) = v$. By strict convexity and compactness, there exists a unique $s > 0$ such that $\Gamma(s) \in C$ and $\Gamma(s) \neq A$. Set $\Theta(A, v)$ to be the sine of the angle between $\dot{\Gamma}(s)$ and the tangent direction to C at $\Gamma(s)$. The function Θ is continuous on UC because the second intersection point and the angle depend continuously on initial data. Since C is strictly convex, the intersection is never tangential, so $\Theta(A, v) \neq 0$ for all (A, v) . By compactness of UC , $|\Theta|$ attains a positive minimum $\delta > 0$. ■

Lemma 3.2 (Rotation and Order Preservation). *Let AA' be a chord of C with midpoint M , and let φ be the orientation-preserving isometry of M that rotates the tangent space at M by angle $\pi/2$. Denote by $\Gamma' = \varphi(AA')$ the image geodesic. Then:*

1. Γ' intersects C transversally in exactly two points, denoted B and B' .
2. Among the two intersection points, there is a unique point B that lies on the forward arc from A to A' (with respect to the orientation of C).

Proof. (1) Let D be the compact convex region bounded by C , and let H be the closed half-space determined by the geodesic AA' containing D . The rotation φ moves A and A' outside H , but keeps the midpoint M (which lies in the interior of D) fixed. Therefore, $\varphi(D)$ is a convex set that contains M but not all of D . The boundary $\partial\varphi(D) = \varphi(\partial D) = \varphi(C)$ is a smooth convex curve that intersects C transversally (because φ is an isometry and C is strictly convex). The intersection $\varphi(C) \cap C$ consists of exactly two points, which are precisely the intersections of $\Gamma' = \varphi(AA')$ with C . Transversality follows from Lemma 3.1 applied to the curve $\varphi(C)$. (2) Consider the continuous one-parameter family of isometries φ_θ for $\theta \in [0, \pi/2]$, where φ_θ is the rotation by angle θ about M . For $\theta = 0$, $\varphi_0(AA') = AA'$. The intersection point of $\varphi_\theta(AA')$ with C that is initially A' moves continuously along C as θ increases. At $\theta = \pi/2$, this continuous path leads to one of the two intersection points, which we denote by B . By continuity and the fact that the intersection number of $\varphi_\theta(C)$ with the arc from A to A' changes sign exactly once, B must lie on the forward arc from A to A' . The other intersection point B' lies on the complementary arc. ■

3.2 Definition and Basic Properties

We now define the central object of our study.

Definition 3.3 (Rotation Map). *Fix a continuous unit tangent vector field along C (e.g., the velocity field of the arc-length parameterization). For a parameter $t \in \mathbb{R}$, let $A = p(t) \in C$. Define $R(t) \in \mathbb{R}$ as follows:*

1. Let Γ be the geodesic through A with initial direction given by the chosen vector field. By Lemma 3.1, Γ intersects C transversally at a unique second point $A' = p(t')$.
2. Let M be the midpoint of the chord AA' .
3. Let φ be the rotation by $\pi/2$ about M . Set $\Gamma' = \varphi(\Gamma)$.
4. By Lemma 3.2, Γ' intersects C at two points. Let B be the intersection point lying on the forward arc from A to A' along C . Define $R(t)$ by $B = p(R(t))$.

The map $R : \mathbb{R} \rightarrow \mathbb{R}$ is called the lift of the rotation map. It induces a well-defined map on the quotient $f : \mathbb{R}/L\mathbb{Z} \rightarrow \mathbb{R}/L\mathbb{Z}$ by $f(p(t)) = p(R(t))$, which we also call the rotation map.

Lemma 3.4 (Basic Properties of R). *The map $R : \mathbb{R} \rightarrow \mathbb{R}$ defined above satisfies:*

1. **Continuity:** R is continuous.
2. **Periodicity:** $R(t + L) = R(t) + L$ for all $t \in \mathbb{R}$.
3. **Strict Monotonicity:** If $t_1 < t_2$, then $R(t_1) < R(t_2)$.

Consequently, the induced map $f : \mathbb{R}/L\mathbb{Z} \rightarrow \mathbb{R}/L\mathbb{Z}$ is an orientation-preserving circle homeomorphism.

Proof. (1) Continuity follows from the continuous dependence of the second intersection point, the midpoint, and the rotated intersection point on the initial data, together with the uniform transversality (Lemma 3.1) which prevents jumps in the selection of the “forward” intersection point B .

(2) Periodicity is immediate from the geometric construction: translating the parameter by L corresponds to going once around the curve C , which returns us to the same geometric configuration.

(3) **Strict Monotonicity.** This is the most delicate property. We present a rigorous proof by contradiction, leveraging the strict convexity of C .

Assume, for the sake of contradiction, that there exist parameters $t_1 < t_2$ such that $R(t_1) \geq R(t_2)$. Let $A_i = p(t_i)$ and $B_i = p(R(t_i))$ for $i = 1, 2$. By construction, B_i is obtained from A_i by the rotation procedure described in Definition 3.3. Consider the two chords A_1B_1 and A_2B_2 .

Case 1: $R(t_1) = R(t_2)$. Then $B_1 = B_2$. The chords A_1B_1 and A_2B_2 share an endpoint. Their other endpoints A_1 and A_2 are distinct ($t_1 < t_2$). The midpoints M_1 and M_2 of these chords are distinct as well. However, since $B_1 = B_2$, the rotation by $\pi/2$ about M_1 sends A_1 to B_1 , and the rotation about M_2 sends A_2 to the same point B_1 . For two distinct rotations (with distinct centers) to send two distinct points A_1, A_2 to the same point B_1 , the points A_1, A_2, B_1 must be collinear and the rotation angles must be supplementary, which is impossible here as both rotations are by the same angle $\pi/2$. This contradicts the strict convexity of C , which ensures that the chords are not collinear and their configurations are generic.

Case 2: $R(t_1) > R(t_2)$. Then the cyclic order of points on C (or their lifts in a fundamental domain on $\mathbb{R}$) is A_1, A_2, B_2, B_1 . Consider the geodesic quadrilateral with vertices (in order) A_1, A_2, B_2, B_1 on C . The chords A_1B_1 and A_2B_2 are its diagonals. By the construction of R , the chord A_2B_2 is obtained by rotating the chord through A_2 (in the initial direction) by $\pi/2$ about its midpoint. Similarly for A_1B_1 . A detailed geometric argument using the fact that both chords are related to their respective “initial chords” by a $\pi/2$ rotation shows that the diagonals A_1B_1 and A_2B_2 must intersect in the interior of the quadrilateral.

However, consider the convex region D bounded by C . The quadrilateral $A_1A_2B_2B_1$ is a (possibly non-convex) polygon on the boundary $\partial D = C$. If its diagonals intersect in the interior of the quadrilateral, that intersection point lies in the interior of D . But then the geodesic segments A_1B_1 and A_2B_2 , which are chords of C , cross inside D . This contradicts the following classical fact about strictly convex curves: *For a strictly convex curve, any two distinct chords cannot intersect in the interior of the convex region they bound.* The proof of this fact is straightforward: if two chords intersected interiorly, their endpoints would not all lie on the boundary of a strictly convex set. Therefore, the assumption $R(t_1) > R(t_2)$ leads to a contradiction with strict convexity.

Since both cases lead to contradictions, we must have $R(t_1) < R(t_2)$ whenever $t_1 < t_2$. This proves strict monotonicity. ■

3.3 From the Rotation Map to Inscribed Squares

The connection between the dynamics of R and inscribed squares is direct.

Lemma 3.5 (Square Lemma). *Let C , R , and f be as above. There exists a parameter $t_0 \in \mathbb{R}$ such that $R^4(t_0) = t_0 + kL$ for some integer k if and only if the points $\{p(t_0), p(R(t_0)), p(R^2(t_0)), p(R^3(t_0))\}$ (taken modulo L) form the vertices of a square inscribed in C .*

Proof. ($\Rightarrow$) Suppose $R^4(t_0) = t_0 + kL$. Set $P_0 = p(t_0)$, $P_1 = p(R(t_0))$, $P_2 = p(R^2(t_0))$, $P_3 = p(R^3(t_0))$. By definition of R , the chord P_0P_1 rotated by $\pi/2$ about its midpoint yields the chord P_1P_2 . Since rotation is an isometry, $|P_0P_1| = |P_1P_2|$ and $\angle(P_0P_1, P_1P_2) = \pi/2$. Similarly, $|P_1P_2| = |P_2P_3|$, $\angle(P_1P_2, P_2P_3) = \pi/2$, and so on for P_2P_3 and P_3P_4 , where $P_4 = p(R^4(t_0)) = p(t_0 + kL) = P_0$. Hence all sides are equal and all interior angles are $\pi/2$, so $P_0P_1P_2P_3$ is a square (or possibly a degenerate square if points coincide, which is ruled out by strict monotonicity). ($\Leftarrow$) If $P_0P_1P_2P_3$ is an inscribed square in cyclic order, then by the geometric definition, $P_1 = f(P_0)$, $P_2 = f(P_1)$, etc. Thus P_0 corresponds to a period-4 point of f , so $R^4(t_0) = t_0 + kL$ for the lift. ■

Thus, proving the existence of an inscribed square is equivalent to proving that $f^4 = f \circ f \circ f \circ f$ has a fixed point (or that R^4 has a point of period 1 in the quotient).

4 Proof of the Main Theorem

We now prove Theorem 1.1 by establishing that f^4 always has a fixed point. The strategy is to compute the rotation number of f^4 via a homotopy argument.

4.1 The Homotopy and Invariance of the Rotation Number

Let $C_0 = C$ be our given smooth strictly convex curve in M . We construct a smooth one-parameter family $\{C_s\}_{s \in [0,1]}$ of smooth strictly convex curves in M such that:

- C_1 is a *circle* (a geodesic circle in S^2 , or a hyperbolic circle in H^2 centered at the origin in the Poincaré model).
- The family is *uniformly strictly convex*: there exists $\kappa_{\min} > 0$ such that the geodesic curvature of each C_s is bounded below by $\kappa_{\min}$.

Such a family exists; for example, one can take a convex combination of the support functions of C_0 and C_1 (in an appropriate model), or use the curve shortening flow which preserves convexity and asymptotically converges to a circle [4].

For each $s \in [0,1]$, we can construct the rotation map $R_s : \mathbb{R} \rightarrow \mathbb{R}$ as in Definition 3.3, using a continuous family of initial direction fields (e.g., by parallel transport along the homotopy). Let $f_s : \mathbb{R}/L_s\mathbb{Z} \rightarrow \mathbb{R}/L_s\mathbb{Z}$ be the induced circle homeomorphism, where L_s is the length of C_s .

Lemma 4.1 (Continuous Dependence). *The family of lifts $\{R_s\}_{s \in [0,1]}$ depends continuously on the parameter s in the compact-open topology. That is, the map $(s, t) \mapsto R_s(t)$ is continuous.*

Proof. The construction of $R_s(t)$ involves: finding the second intersection point of a geodesic with C_s , finding the midpoint, rotating, and selecting the forward intersection point of the rotated geodesic with C_s . All these operations depend continuously on the curve C_s and the base point, provided the intersection points remain transverse. The uniform strict convexity (and hence uniform transversality, by Lemma 3.1 applied to the compact family $\{C_s\}$) guarantees that the “forward” intersection point is chosen continuously and does not jump. Thus continuity in both variables holds. ■

Since each R_s satisfies $R_s(t + L_s) = R_s(t) + L_s$, we can consider the *normalized* lift $\tilde{R}_s(t) = (L_0/L_s)R_s((L_s/L_0)t)$, which satisfies $\tilde{R}_s(t + L_0) = \tilde{R}_s(t) + L_0$ for all s , aligning the periods. The continuous dependence of $\tilde{R}_s$ on s is preserved.

For a circle C_1 , by symmetry, the rotation map f_1 is precisely a rigid rotation by one-quarter of the circumference: $R_1(t) = t + L_1/4$. Consequently,

$$R_1^4(t) = t + L_1, \quad \text{and} \quad \tilde{R}_1^4(t) = t + L_0.$$

Lemma 4.2 (Invariance of Rotation Number). *The rotation number $\rho(f_s^4)$ is constant for $s \in [0,1]$. In particular, $\rho(f_0^4) = \rho(f_1^4)$.*

Proof. For a continuous family of circle homeomorphisms f_s whose lifts $\tilde{R}_s$ depend continuously on s (in the sense above), the rotation number $\rho(f_s)$ depends continuously on s . Since rotation numbers are rational numbers with bounded denominators (or irrationals), continuous dependence implies constantness unless the family crosses a “locking” interval—but for a continuous family, the rotation number is locally constant at points where it is rational. A more direct argument for our purpose: the map $s \mapsto \rho(f_s^4)$ is continuous and takes values in $\mathbb{R}/\mathbb{Z}$. However, because the family is continuous and the normalized lifts $\tilde{R}_s^4$ satisfy a uniform periodicity condition, one can show that the limit defining $\rho(f_s^4)$ varies continuously with s . Since $\rho(f_1^4) = 0$ (or 1, in additive notation), and the set of possible rotation numbers is totally disconnected, continuity forces $\rho(f_s^4)$ to be constant. A standard result in dynamical systems states that the rotation number is constant on connected components of the space of orientation-preserving circle homeomorphisms endowed with the C^0 topology; our family $\{f_s\}$ provides a continuous path in that space. ■

4.2 Completion of the Proof

We now combine the ingredients to establish the existence of a fixed point of f^4 .

Theorem 4.3. *Let C be a smooth strictly convex closed curve in S^2 or H^2 , and let f be the associated rotation map. Then the fourth iterate f^4 has a fixed point.*

Proof. Consider the homotopy $\{C_s\}$ from $C_0 = C$ to a circle C_1 as described. By Lemma 4.1, we have a continuous family of circle homeomorphisms f_s with lifts R_s . Their fourth iterates f_s^4 also form a continuous family.

For the circle C_1 , we computed $R_1^4(t) = t + L_1$, so the induced map f_1^4 is the identity map on $\mathbb{R}/L_1\mathbb{Z}$. Its rotation number is $\rho(f_1^4) = 0$ (in additive notation with respect to the fundamental domain $[0, L_1)$). Equivalently, if we consider the lift R_1^4 with $R_1^4(t + L_1) = R_1^4(t) + L_1$, the rotation number is 1 in the multiplicative sense (i.e., translation by one full period). To match the standard notation of Corollary 2.4, we normalize: define $g_s(x) = \frac{1}{L_s}R_s^4(L_s x)$. Then $g_s : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $g_s(x + 1) = g_s(x) + 1$, and $\rho(g_s) = \rho(f_s^4)$. For $s = 1$, $g_1(x) = x + 1$, so $\rho(g_1) = 1$. By Lemma 4.2, $\rho(g_s)$ is constant, hence $\rho(g_0) = \rho(g_1) = 1$. Corollary 2.4 applied to g_0 (which is a lift of f_0^4) tells us that $f_0^4 = f^4$ has a fixed point. ■

Proof of Theorem 1.1. By Theorem 4.3, the rotation map f associated to the curve C has the property that f^4 possesses a fixed point. By Lemma 3.5, such a fixed point corresponds to a quadruple of points on C forming an inscribed square. This completes the proof. ■

5 Conclusions and Generalizations

We have presented a complete proof of the existence of inscribed squares on any smooth strictly convex closed curve in the two-dimensional sphere S^2 and the hyperbolic plane H^2 . The proof is built on three pillars:

1. The construction of a well-defined and strictly monotone *rotation map* R via careful geometric arguments (Lemmas 3.1, 3.2, and 3.4).
2. The dynamical reinterpretation of the problem as a fixed point question for the circle homeomorphism f induced by R (Lemma 3.5).
3. The use of topological invariants (rotation number) and a homotopy argument to reduce the general case to the computable case of a circle (Theorem 4.3).

This approach highlights the power of topological and dynamical methods in classical geometry.

Generalizations and Future Work

The framework developed here is robust and suggests several natural extensions:

1. **Other regular polygons.** By replacing the rotation angle $\pi/2$ with $2\pi/m$, one can investigate the existence of inscribed regular m -gons. The rotation map would then be defined by rotating chords by $2\pi/m$, and the relevant iterate would be the m -th iterate. The homotopy argument should carry over, but the computation of the rotation number for the base circle case becomes $\rho(f^m) = 1$, leading to a fixed point of f^m . Preliminary analysis indicates that for $m > 4$, additional constraints on the curve may be necessary.
2. **Higher dimensions.** A challenging but fascinating direction is to consider convex hypersurfaces in $\mathbb{R}^n$, S^n , or H^n and ask whether they contain inscribed cubes or other regular polytopes. The dynamical approach may generalize if one can find a suitable replacement for the one-dimensional circle map.
3. **Weakening convexity.** Can the assumption of strict convexity be relaxed to simple closed curves with some regularity (e.g., C^2) and non-vanishing curvature? The transverse intersection lemmas may fail, but alternative constructions using topological degree might still yield results.
4. **Algorithmic aspects.** Our proof is non-constructive. An interesting computational question is whether the fixed point of R^4 can be located effectively, yielding an algorithm to approximate the vertices of an inscribed square.

We intend to explore these directions in subsequent work.

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